Explainable and Robust Deep Learning for Anomaly Detection in Industrial Control Systems: A WDNN–GRU Comparison

Aaditya Singh[1*], Sreeya Anne[1] and Mudit Punetha[1]

[1*]Department of Computer Science and Engineering, Bennett University, Greater Noida, Uttar Pradesh, India.

*Corresponding author(s). E-mail(s): S24CSEU0616@bennett.edu.in;
Contributing authors: S24CSEU0562@bennett.edu.in;
S24CSEU0597@bennett.edu.in;

Abstract

Industrial control systems (ICSs) require anomaly detectors that distinguish malicious process deviations from ordinary operating variations. This work evaluates a DAICS-inspired wide-and-deep neural network (WDNN) baseline against an independently trained gated recurrent unit (GRU) predictor using a processed Secure Water Treatment (SWaT) dataset. Both predictors use the same 44 process variables, forecast four future readings for each of 24 sensors, and are evaluated with model-specific threshold-tuning networks and actuator-state checking. We add post-hoc sensor-level residual attribution and a paired Gaussian sensor-noise robustness study. On 348,537 pooled clean test windows, GRU achieves precision 0.6675, recall 0.8261, F1 0.7383, and false-positive rate (FPR) 0.0763; WDNN achieves precision 0.5153, recall 0.8818, F1 0.6505, and FPR 0.1538. GRU retains higher pooled F1 and lower FPR under the evaluated perturbations, while WDNN retains higher attack recall. On 148 matched clean detection episodes, the models agree on the highest-ranked residual-contributing sensor 96.6% of the time. The study is an adapted DAICS comparison rather than an exact reproduction: original online model adaptation is not implemented, the TTNN training target is approximated, and available labels do not support reliable per-attack delay measurement.

Keywords: Industrial control systems, anomaly detection, SWaT, wide-and-deep neural network, gated recurrent unit, explainability, robustness

1 Introduction

Industrial control systems coordinate sensors, actuators, and programmable logic controllers in safety-sensitive processes such as water treatment and manufacturing. Their telemetry provides an opportunity to identify attacks or equipment faults through departures from learned normal behavior. Prediction-based detectors learn process dynamics from normal records and identify unusually large deviations between expected and observed sensor states. However, false alarms may arise from environmental noise and shifts in otherwise legitimate operating conditions [1, 2].

The original DAICS framework combines a wide-and-deep sensor predictor with a database of allowed actuator states, dynamic error thresholds, and procedures for updating its model when a technician confirms a false alarm [1]. Recurrent models such as the gated recurrent unit (GRU) offer another way to model temporal dependencies [3]. The relative false-alarm and attack-detection trade-offs of a compact GRU alternative under identical processing and noise conditions are therefore worth examining.

This project makes four evaluated contributions: (i) a DAICS-inspired WDNN implementation on an explicitly documented 44-variable SWaT subset; (ii) an independent, similarly evaluated GRU forecasting alternative; (iii) sensor-level residual attribution for detected section episodes; and (iv) a paired robustness comparison based on shared, reproducible noise arrays. Unlike the original DAICS work, this implementation does not perform online TTNN weight updates or few-time-steps model adaptation. Our results should thus be interpreted as a *fixed-model comparison* rather than a full adaptive-DAICS reproduction.

2 Related Work

2.1 Prediction-based ICS anomaly detection

Goh *et al.* studied recurrent prediction for anomaly detection in cyber-physical systems, using cumulative-error monitoring to recognize deviations from normal operation [2]. The advantage of recurrent prediction is its temporal representation, but the selection and maintenance of error limits can complicate deployment. Kravchik and Shabtai evaluated convolutional networks for SWaT anomaly detection and measured deviations between predicted and observed readings [4]. Their findings demonstrate that temporal convolution is a strong alternative to recurrent forecasting rather than assuming an RNN is always superior.

Abdelaty *et al.* developed AADS as a noise-robust detector and subsequently introduced DAICS, which combines wide/deep prediction, actuator-state checking, threshold tuning, and few-time-steps model adaptation [1, 5]. Its wide branch is intended to memorize feature co-occurrences, while the deep branch generalizes nonlinear relationships. DAICS also evaluates additive noise and domain shift; therefore Gaussian perturbation testing itself is not a new method introduced in this project. We instead use a *paired, fixed-reference* comparison of two predictors, plus residual-based explanation of their alarms.

2.2 Temporal representation and interpretation

The GRU introduced in encoder–decoder research uses gating to retain or reset information across a sequence [3]. In this study, it replaces the forecasting network for a separate experimental alternative rather than being fused into the WDNN. Residual attribution answers a narrower question than feature-level causal explanation: it identifies which forecasted sensors account for the observed squared error, not which physical sensor or attack caused the incident.

3 Problem Formulation

Let $x_t \in \mathbb{R}^{44}$ denote the measurements at one second: 24 scaled continuous sensor channels and 20 unscaled discrete actuator channels. For an input start index i, both predictors receive

$$X_i = [x_i, \ldots, x_{i+W_{\text{in}}-1}], \quad W_{\text{in}} = 60. \tag{1}$$

A horizon $H = 50$ separates the input from the first predicted row. Four target samples begin at $i + W_{\text{in}} + H = i + 110$, giving $W_{\text{out}} = 4$. The forecast targets consist of the 24 sensor channels, divided into three sections of 5, 7, and 12 sensors.

For output section g containing m_g sensors, let $\hat{y}_{i,g,j,h}$ and $y_{i,g,j,h}$ be the normalized predicted and observed readings for sensor j at forecast step h. The section forecast error is

$$E_{i,g} = \frac{1}{m_g W_{\text{out}}} \sum_{j=1}^{m_g} \sum_{h=1}^{W_{\text{out}}} (y_{i,g,j,h} - \hat{y}_{i,g,j,h})^2. \tag{2}$$

The training objective is the sum of the three section-wise mean squared errors. Detection combines thresholded section errors after a persistence rule with a lookup for unseen actuator-state combinations.

4 Methodology

4.1 Data and preprocessing

The source exports contained 1,387,098 normal rows and 54,621 attack-source rows. Removing 495,000 exact duplicate normal records yielded 892,098 unique normal rows. Seven extensively unavailable process variables (MV101, AIT201, MV201, P201, P202, P204, and MV303) were excluded, leaving 24 sensors and 20 actuators. Sensor MinMax scaling was fitted on training records only. The original DAICS SWaT configuration used 51 variables; our results refer specifically to the 44-variable processed configuration [1, 6].

The first seven days of normal data were divided chronologically into 478,365 training and 119,591 benign validation records. A further 294,142 normal records and the separate 54,621-record attack source were independently windowed for testing (Table 1). All attack-source rows have the Attack label. Although attack timestamps overlap the calendar dates of normal validation, no attack observations enter model fitting, normalization, or threshold calibration. The split is thus source-isolated rather than one global chronological cutoff.

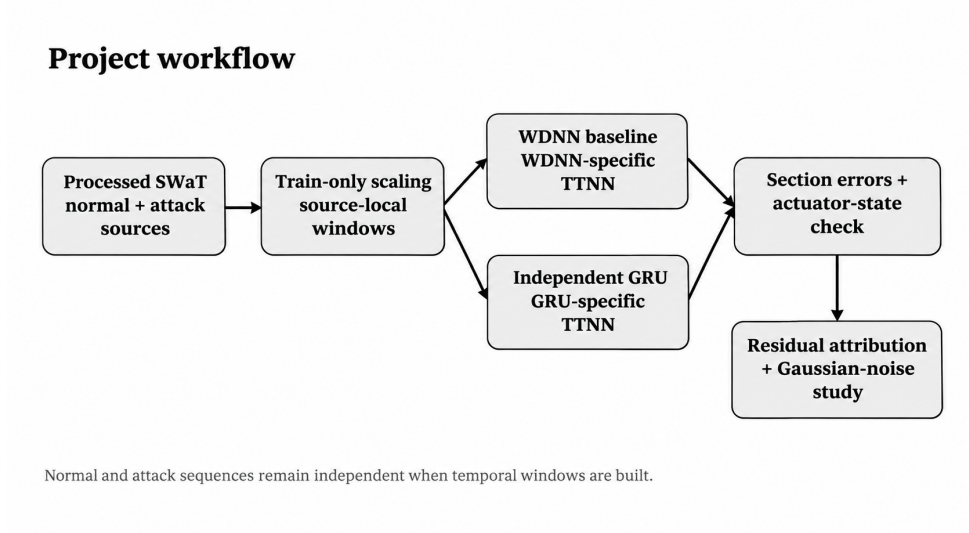


Fig. 1 Comparative processing and evaluation workflow. WDNN and GRU are separate predictor alternatives; explainability operates after detection, while robustness is evaluated with shared sensor perturbations.

Table 1 Processed Data Splits

Source	Rows	Windows
Train normal	478,365	478,252
Validation normal	119,591	119,478
Test normal	294,142	294,029
Test attack	54,621	54,508

4.2 DAICS-inspired WDNN baseline

The WDNN comprises a wide branch using a flattened dense projection, and a deep branch with dense, Conv1D, max-pooling, and fusion layers (Fig. 3). The deep path uses 180 dense units, successive convolution stages of 64 and 128 filters, and a subsequent dense representation; the fused representation feeds three independent section heads. Hidden layers use LeakyReLU with negative slope 0.01. Importantly, the wide projection does *not* explicitly enumerate the feature cross-products proposed by the original paper [1]; it is a documented approximation. The training run uses SGD, learning rate 0.01, batch size 32, and 100 epochs.

4.3 Independent GRU predictor

The GRU processes each 44×60 input as a 60-step sequence of 44-dimensional feature vectors. It uses one GRU layer with 64 hidden units. The last hidden state feeds three independent linear heads producing tensors of shapes $(B, 5, 4)$, $(B, 7, 4)$, and $(B, 12, 4)$, where B is the batch size (Fig. 4). The model contains 27,360 parameters

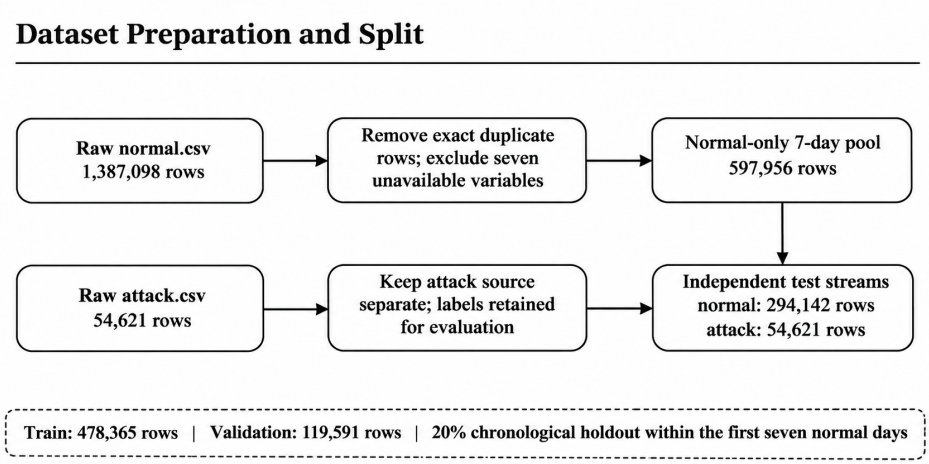


Fig. 2 Cleaning and source-isolated data partitioning. Training-only normalization is applied after splitting.

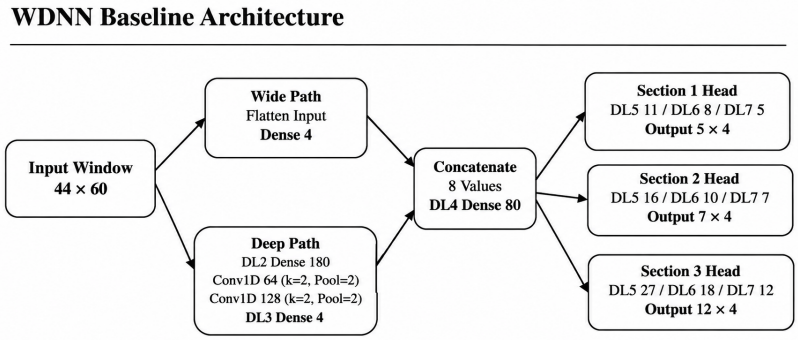


Fig. 3 Implemented WDNN with linear wide projection, convolutional deep path, fusion, and output sections of 5, 7, and 12 sensors. This reflects the project implementation rather than an exact copy of DAICS Fig. 2.

and is trained with Adam (learning rate 0.001), batch size 32, gradient clipping 1.0, and a 15-epoch maximum with validation early stopping patience 3. The epoch-12 checkpoint has the best recorded GRU validation loss of 0.0101601.

4.4 Thresholds, persistence, and actuator states

Each predictor is separately calibrated on benign validation forecast errors. The base threshold is $T_{\text{base}} = \mu_{\text{val}} + \sigma_{\text{val}}$. Three threshold-tuning neural networks (TTNNs) use chronological historical MSE windows of length 60; the project uses a 59-sample median filter with reflected boundary handling. A threshold for section g takes the

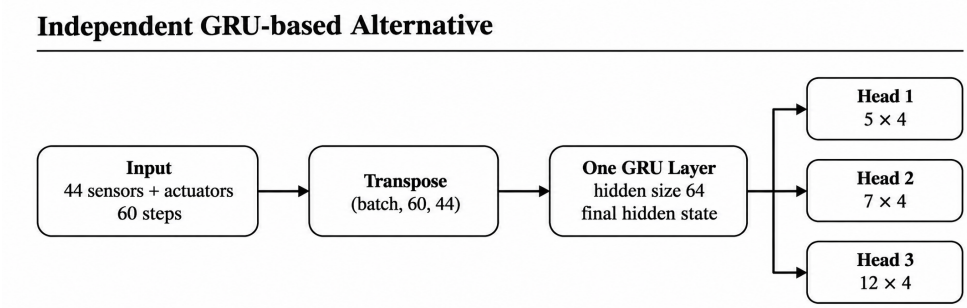


Fig. 4 Single-layer GRU predictor with three independently ordered four-step sensor heads.

form

$$T_g(t) = T_{\text{base}} + \max\{\widehat{E}_{g,\text{historical batch}}\}. \tag{3}$$

A section alert requires errors above its threshold for $W_{\text{anom}} = 30$ consecutive samples; an unseen discrete actuator tuple also triggers an alarm, and the two paths are combined by OR. Historical-only alignment was verified in the baseline smoke tests. The normal-state actuator database contains 70 observed combinations, not the 146 reported in the original DAICS work.

TTNN training uses an explicitly documented *next-MSE regression approximation*: the original paper does not fully specify a conventional supervised target or loss for online updates. In particular, the original TTNN online SGD weight update and few-time-steps WDNN adaptation are *not* reproduced. During the robustness experiments, even the historical TTNN threshold trajectories are frozen from clean-reference source-local errors to isolate input-noise effects.

4.5 Residual-based explainability

For sensor j in section g, the horizon-averaged squared residual and its percentage share of section error are

$$r_{i,g,j} = \frac{1}{W_{\text{out}}} \sum_{h=1}^{W_{\text{out}}} (y_{i,g,j,h} - \hat{y}_{i,g,j,h})^2, \tag{4}$$

$$C_{i,g,j} = 100 \, \frac{r_{i,g,j}}{\sum_k r_{i,g,k}}, \tag{5}$$

with zero contributions assigned when the section sum is zero. For detected section episodes, the five highest shares are stored with forecast timestamps and the separate alarm-confirmation timestamp. These values are in training-normalized sensor units. High contributions indicate residual magnitude, not causal responsibility.

4.6 Controlled noise robustness protocol

The existing test sensor readings were perturbed before window generation according to $x'_{t,j} = x_{t,j} + \epsilon_{t,j}$ with $\epsilon_{t,j} \sim \mathcal{N}(0, (\lambda\sigma_j)^2)$, where σ_j is the training-set standard

Table 2 Clean Pooled Point-Level Performance

Model	Prec.	Recall	F_1	FPR
WDNN	0.5153	**0.8818**	0.6505	0.1538
GRU	**0.6675**	0.8261	**0.7383**	**0.0763**

deviation of normalized sensor j, and $\lambda \in \{0, 0.01, 0.05, 0.10\}$. Actuator states and test labels were unchanged; readings were not clipped. The same stored noise arrays were applied to both models, and each model retained its own clean-reference, historical threshold trajectory for all conditions. This is not a deployment simulation of TTNN adaptation to noisy operating history.

5 Experimental Setup

Experiments were performed with Python 3.12.10 and PyTorch 2.5.1+cu121 on an NVIDIA GeForce RTX 3050 Laptop GPU with 6 GB VRAM. On-demand temporal windows avoided materializing the full overlapping tensor dataset. The WDNN run evaluated the saved epoch-100 checkpoint (validation loss 0.0130859), although the lowest logged validation loss, 0.0119309, occurred at epoch 90 and those weights were unavailable. GRU selected its lowest-validation-loss epoch-12 checkpoint. Therefore checkpoint selection differs between the two models.

We use point-level precision $P = TP/(TP + FP)$, recall $R = TP/(TP + FN)$, $F_1 = 2PR/(P+R)$ and false-positive rate $FPR = FP/(FP+TN)$. Each test window has the label of its forecast interval. Normal and attack test sources were evaluated separately before pooling. The attack source contains one contiguous all-Attack label interval; consequently, the stored zero-second delay does not measure detection delay for the 36 distinct attack episodes in the original SWaT study [1, 6].

6 Results and Discussion

6.1 Training behavior

Figure 5 presents the saved loss histories. WDNN uses 100 training epochs and its final checkpoint; GRU uses a 15-epoch run and the selected epoch-12 checkpoint. Different optimizers and architectures mean their loss trajectories and epoch counts should not be treated as a controlled comparison of optimization efficiency.

6.2 Clean detection performance

Table 2 and Fig. 6 show that GRU yields a higher pooled F1 and lower FPR than WDNN. Specifically, GRU reduces false positives from 45,214 to 22,434, but loses 3,039 correctly detected attack windows. The lower false-alarm rate is therefore accompanied by reduced attack recall. The figures are comparisons of the *saved* epoch-100 WDNN and best-validation epoch-12 GRU, not a controlled best-checkpoint comparison.

On the normal-only test source, FPR is 0.1538 for WDNN and 0.0763 for GRU. Its positive-class F1 is not informative because no windows carry an Attack label.

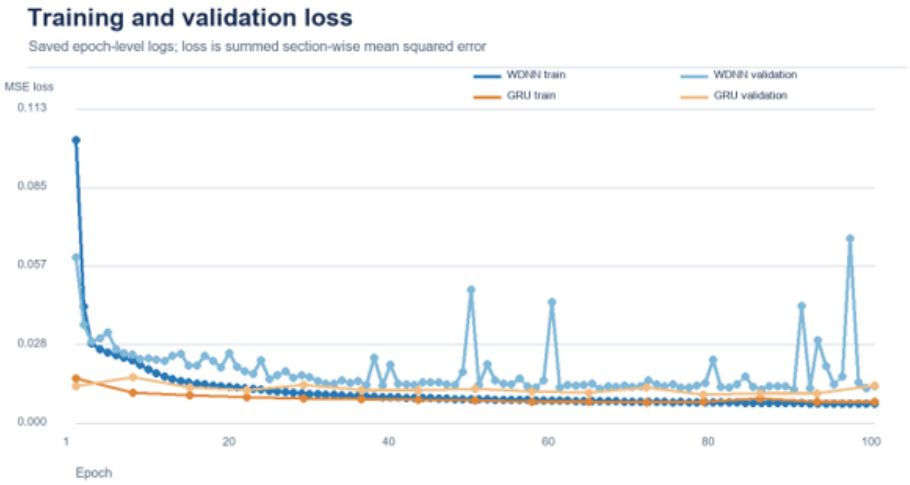


Fig. 5 Recorded training and benign-validation losses for WDNN and GRU. Model settings and best-checkpoint selection procedures differ.

Table 3 Clean Pooled Confusion Counts

Model	TP	FP	TN	FN
WDNN	48,067	45,214	248,815	6,441
GRU	45,028	22,434	271,595	9,480

The separate attack-only source yields recall 0.8818 for WDNN and 0.8261 for GRU; by construction that source has no negative labels. Reporting both source-specific behavior and pooled results avoids confusing attack detection with false-alarm control.

6.3 Noise robustness

The robustness comparison in Table 4, Fig. 7, and Fig. 8 uses the same saved perturbation samples in both models, with model-specific clean historical thresholds held fixed. GRU retains higher F1 and lower FPR under all conditions, whereas WDNN has higher attack recall.

At the high-noise level, WDNN false positives decrease from 45,214 to 43,877 while attack true positives decrease from 48,067 to 47,991. For GRU, false positives decrease from 22,434 to 19,476 while true positives decrease from 45,028 to 44,906. Thus, although pooled F1 rises modestly, attack recall declines for both models. This result must not be interpreted as a general benefit of adding noise or as evidence of on-line noise adaptation: the test uses one saved perturbation realization per condition and frozen clean-reference thresholds.

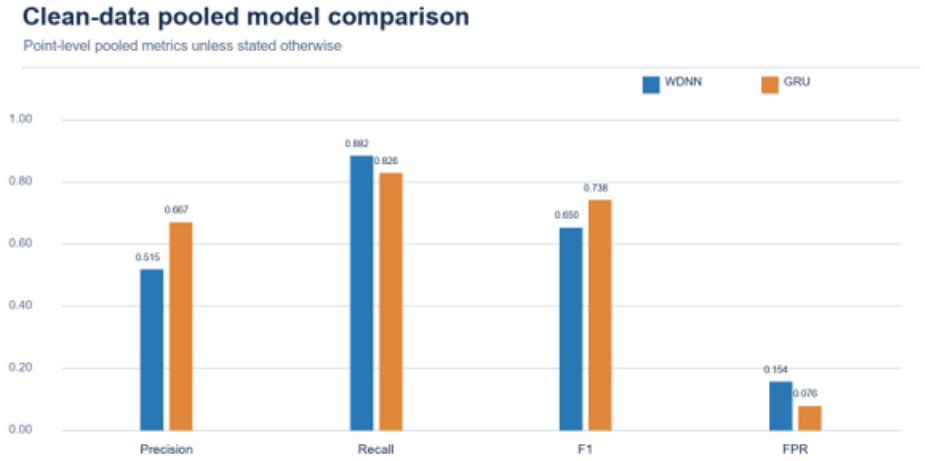


Fig. 6 Clean pooled precision, recall, F1 and false-positive rate for the two evaluated checkpoints.

Table 4 Pooled Robustness Results Under Identical Sensor Perturbations

Noise	Model	Precision	Recall	F_1	FPR
Clean	WDNN	0.5153	0.8818	0.6505	0.1538
Clean	GRU	0.6675	0.8261	0.7383	0.0763
Low (0.01σ)	WDNN	0.5156	0.8818	0.6507	0.1536
Low (0.01σ)	GRU	0.6670	0.8260	0.7380	0.0764
Medium (0.05σ)	WDNN	0.5178	0.8813	0.6523	0.1521
Medium (0.05σ)	GRU	0.6796	0.8251	0.7453	0.0721
High (0.10σ)	WDNN	0.5224	0.8804	0.6557	0.1492
High (0.10σ)	GRU	0.6975	0.8238	0.7554	0.0662

6.4 Residual attribution and consistency

On clean data, WDNN produced 637 detected sensor-section episodes and GRU produced 469. Only 148 episodes were matched; 489 were WDNN-only and 321 GRU-only. For those matched episodes, the rank-one contributing sensor agreed 96.6% of the time and the mean top-five sensor-set Jaccard similarity was 0.782. These statistics are conditional on matched alarms and must not be generalized to all detected events. Frequently highest-ranked sensors included DPIT301 and FIT201 (Fig. 9).

Figure 10 illustrates an explanation associated with a confirmed alarm. The displayed contribution quantifies forecast mismatch, not causal attack origin. Under noise, rank-one agreement on matched alarms ranged approximately from 96.6% to 97.9%, while top-five Jaccard similarity ranged from 0.772 to 0.796. Since the models detected substantially different episodes, agreement alone does not establish equivalent sensitivity or explainability quality.

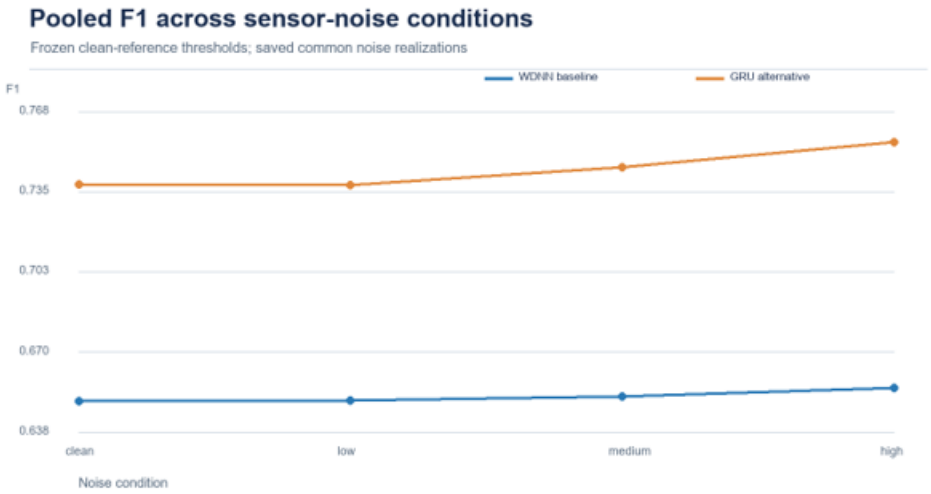


Fig. 7 Pooled F1 across clean, low, medium, and high sensor-noise conditions.

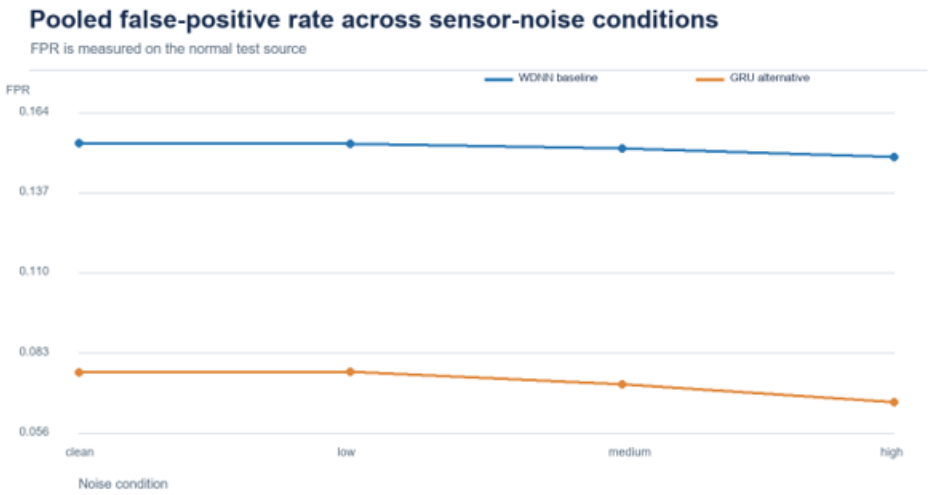


Fig. 8 Pooled FPR across the identical noise conditions.

7 Limitations and Future Work

The original DAICS paper describes 51 SWaT process variables, PLC-associated sections, cross-product wide transformations, online TTNN updates, and few-time-steps adaptations [1]. This project uses 44 retained variables, approximate stage-based sensor groups, a dense-projection wide branch, a next-MSE TTNN training-target

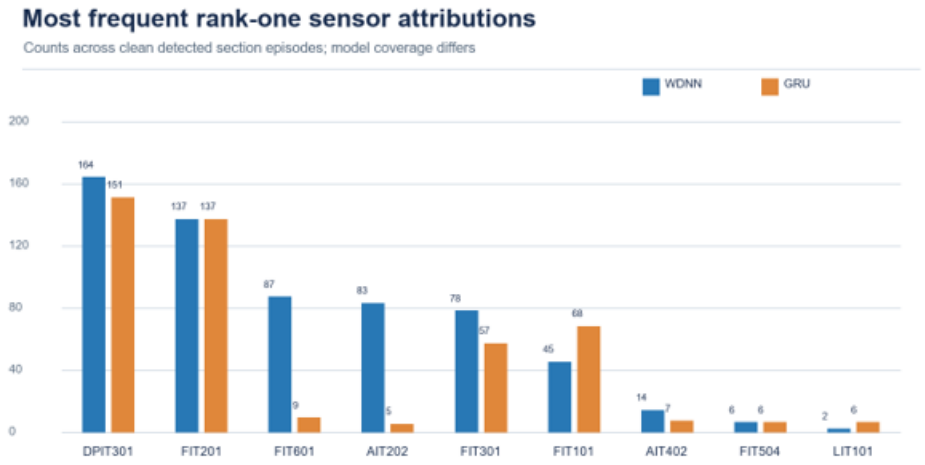


Fig. 9 Frequently highest-ranked residual contributors to clean detected sensor-section episodes.



Fig. 10 Illustrative record from saved GRU explanations, showing ranked normalized-scale sensor residual contributions for a detected section episode.

approximation, frozen TTNN weights at evaluation, and no online model/database updates. The available actuator-state database contains 70 combinations rather than the original paper's 146.

The processed attack source contains one continuous Attack-labelled interval, so a recorded delay of zero seconds does not represent detection delays for the 36 attacks

in the original dataset. Validation timestamps overlap the attack-source calendar period despite source isolation. WDNN test evaluation uses the epoch-100 checkpoint while GRU uses its best epoch-12 checkpoint; this is a comparison of available artifacts, not an equal checkpoint-selection protocol. Attribution is residual accounting rather than a physical causal explanation. The robustness protocol evaluates fixed clean-reference threshold trajectories, not adaptation to noisy histories. Future work should recover event-level attack metadata, implement and validate online adaptation, evaluate alternative threshold targets, and run a best-validation-matched checkpoint comparison.

8 Conclusion

An independently trained, compact GRU predictor achieved higher pooled F1 and approximately half the false-positive rate of an available DAICS-inspired WDNN checkpoint on a processed SWaT test configuration. WDNN, however, detected more attack-labelled windows. Residual attribution identified broadly similar leading sensor-error contributors on the matched alarms, while robustness evaluations indicated stable but non-monotonic precision–recall trade-offs under controlled sensor noise. These conclusions are confined to the saved fixed-model experiments, data preprocessing decisions, and documented DAICS approximations.

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Declarations

Funding: No funding information was provided for this project; the authors should confirm before submission.

Competing interests: The authors should confirm before submission.

Data availability: The experiments use processed SWaT records; dataset access and redistribution permissions should be confirmed against the original SWaT provider's terms.

Code availability: The implementation was developed for the project. Public release details have not been provided.

Author contributions: To be confirmed by the authors before submission.

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